bayanihan-net

A hybrid multi-agent system for disaster-response coordination under scarcity

Designing & Building Agentic AI Systems — Capstone

Marikina flood basin, Metro Manila · reproducible from seed 20260620

A consolidation of the repo's README, eight design docs, and the arXiv-style paper.
Graded on the working code; every number below regenerates from a seed with no API key.

Roadmap

1. The problem (system brief)
2. Why MAS — a single agent is not enough
3. Architecture
4. Agent roster
5. Communication contract
6. Coordination — a defended hybrid
7. Incentives & emergence
8. Interoperability (MCP / A2A)
9. Prototype & simulation
10. Evaluation (four levels, honest result)
11. Safety & governance
12. MARL bridge & reflection

1 The problem — a Marikina typhoon flood

A typhoon stalls over Metro Manila; the Marikina River rises through its alarm levels and six low-lying barangays flood within hours. The city disaster office (CDRRMC) must allocate a **small fleet of scarce assets** across many simultaneous, evolving incidents.

Stakeholders & stakes

- ∼5,100 residents at risk across 6 barangays
- 9 assets: 4 boats, 3 trucks, 2 medical teams
- Duty officer holds authority over irreversible calls
- Partner agencies (mutual aid); upstream feeds (PAGASA/MMDA/DOH)

Why it is hard

- Scarcity is binding (capacity-saturated)
- Irreversibility (a crew on-scene cannot be un-sent)
- Partial observability & degradation
- Adversarial / noisy citizen reports (Sybil)

Objective

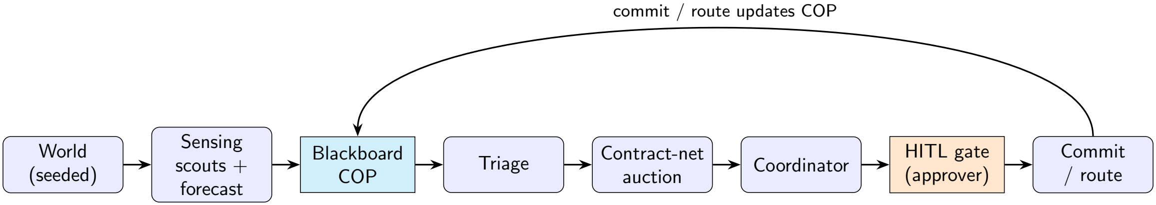
Minimize total **severity-weighted unmet need**, equitably and safely, under hard asset scarcity. (Geography/figures are illustrative stand-ins, not official data.)

2 Why a multi-agent system — a single agent is the wrong shape

1. **Partial observability.** No vantage sees the whole flood — scouts, the river gauge, and citizen reports each see a slice; the truth is *assembled*.
2. **Specialized expertise.** Triage, mass-casualty judgement, logistics, routing under flooding, and hydrology are different competencies with different state and tools.
3. **Parallel decomposition.** Many concurrent incidents; a single agent serializes.
4. **Resilience / graceful degradation.** A typhoon degrades power and comms — a single point of control is a single point of failure, fatal in disaster response.
5. **Organizational alignment.** Real response *is* multi-agency: barangay → city → NDRRMC, plus relief, health, rescue, NGOs.

We adopt MAS because the *problem* is distributed, specialized, concurrent, failure-prone, and inter-organizational — not because MAS is fashionable.

3 Architecture



Cross-cutting: Governance (capability scopes, gate rules, pure) · Auditor (golden signals + emergence metrics, read-only) · Interop (MCP tools, A2A mutual aid)

Each tick: world advances → COP refreshes → triage → auction → award (through the HITL gate) → assets move along risk-aware routes, *validated against the true flood at arrival*. The control plane is deterministic; only the world is stochastic (one seed).

4 Agent roster — roles, scopes, escalation (enforced in code)

Agent	Responsibility	Scopes / escalates
Sensing scout $\times 2$	citizen reports $\rightarrow$ COP incidents; flag Sybil	`cop:write`
Hydromet / forecast	post river level (PAGASA gauge via MCP)	`cop:write`, `tool:pagasa.read`
Triage	prioritize, dedupe, suppress uncorroborated Sybil; announce tasks	`taskboard:write`
Logistics $\times 2$	own & bid scarce assets (rescue boats / relief trucks)	`asset:bid`, `asset:commit`
Medical	bid teams; watch mass-casualty overload	`+escalation:raise` (edge-triggered)
Routing	risk-aware routes over a *belief* of the flood	`tool:mmda.read` (advises)
Coordinator	re-score on welfare, award, gate, atomic commit, rollback	`asset:award`, `rollback:issue`
Human approver	approve/deny irreversible commits from a decision package	final authority
Auditor	read-only: outcome + emergence reports	`audit:read`

Memory is **externalized**: the durable store is the COP + append-only event log, not hidden per-agent state. Boundaries are enforced — every COP write is gated at the bus on the sender's `cop:write` scope.

5 Communication contract

Typed envelope (Pydantic, `extra="forbid"`)

- `schema_version`, `trace_id`, `message_id`
- `correlation_id`, `idempotency_key`, `priority`
- `tick`, `deadline_tick`, `security_context`, `payload`

`message_id` is deterministic (`agent:tick:seq:type`) — **no uuid4**, so a seed replays the exact message sequence. 12 message verbs; a test asserts every verb has a payload.

Routing

- pub/sub on the blackboard (observations)
- directed contract-net (CFP → bid → award)
- dedicated escalation channel (no-drop path)

Shared state

- idempotency, two layers (transport + content-key fusion)
- leases / freshness (stale-COP guard)
- atomic `try_commit` (the no-double-commit chokepoint)

The MAS failure catalogue (deadlock, message storm, duplicates, stale state) is mapped row-by-row to a mitigation; lossy comms and Sybil are handled as domain threats.

6 Coordination — a defended hybrid

The mechanism

Blackboard (shared COP) + **contract-net** (scarce-asset auction) + **supervisor with HITL** (accountable arbitration & escalation). Each leg is the right primitive for one sub-problem.

Why not a pure mechanism (each option on the course's coordination menu was considered):

- *Pure supervisor* — a single point of failure (fatal here) and a bottleneck.
- *Pure market / contract-net* — efficient allocation but *no shared awareness* and no accountable escalation.
- *Pure consensus* — too slow and expensive for time-critical life-safety decisions.
- *Pure blackboard* — shared awareness but *no decision-maker*.

The hybrid composes all three: shared awareness *and* efficient allocation *and* accountable escalation — and it **degrades gracefully** (lose the coordinator → agents fall back on the last-known COP).

7 Incentives & emergence

Incentives — local vs. global
Each logistics agent locally wants its own utilization; the global goal is to minimize severity-weighted harm. Resolved *structurally*: **the bidder does not choose the winner**. The coordinator ranks on a global welfare score

$$W = \frac{\mathsf{sev} \cdot \min(\mathsf{cap}, \mathsf{ppl})}{\mathsf{eta} + 1}\left(1 - \tfrac{1}{2}\,\mathsf{risk}\right)$$

so hoarding or cherry-picking cannot win — the welfare-maximizing (VCG-flavoured, no payments) allocation rule.

Emergence — named, with detector → guardrail
Useful: bayanihan division of labour; adaptive routing; load spreading (entropy ≈ 0.97).

Unwanted:

- oscillation → reassignment rate → hysteresis
- inequity → coverage Gini / worst-served → severity-first
- report-storm → Shannon entropy → content-key dedup
- stale-COP → freshness → leases + arrival validation
- over-escalation → rate → edge-triggered

8 Interoperability — MCP and A2A boundaries

Two distinct boundaries, both **real in code** with mocked upstreams.

MCP — scoped tool/context

- PAGASA river gauge: the **live** hop, invoked every tick (**48 traced calls/run**)
- MMDA closures, DOH beds: *registered & scope-declared but not invoked*
- every adapter declares its scope; an under-scoped caller is refused (tested)

A2A — cross-org work exchange

- medical mutual aid from a neighbouring agency (Quezon City CDRRMC)
- typed task; **local access policy owned by the receiver**
- idempotency + retry; trace propagation (1 request/run, completed)

The *boundary and the policy gate* are real in code — the part that matters for safety — not an LLM tool-call layer; the upstream services are simulated.

9 Prototype — a deterministic, reproducible simulation

Discrete-event simulation over a **12-hour horizon** (48 ticks $\times$ 15 min), seed 20260620. One `cli run` produces a full worked scenario and stamped evidence.

Committed evidence

- `run_log.txt` — one incident, end to end
- `audit_log.jsonl` — immutable event spine
- `decision_package.json` — a real HITL package
- `scenario_report.json` — outcome + golden signals
- 5 figures (COP timeline, coverage, RL curve, . . .)

Worked run (seed 20260620)

- served **266 / 776** people (34.3%)
- dispatch SLA 47.0%; 1 escalation
- HITL: 12 approvals / 7 denials
- **byte-identical** on re-run; fingerprint `013f8d40ba7f`

Proven from a clean room: a fresh checkout + the README quickstart regenerates all 8 evidence artifacts *byte-for-byte*.

10 Evaluation — four levels, and an honest result

Two tiers: **Tier A** invariants are a hard gate; **Tier B** is the comparative study (12 paired seeds, 5 ablation baselines, means ±SE). A delta is flagged on a 2-SE *screen* (not a *t*-test — stated honestly). Every level emits real numbers (agent / interaction / system / human+emergence) plus the six MAS golden signals.

Policy	sev-wt served	worst-served
hybrid	0.353	0.220
no_governance	0.355	0.222
greedy_nearest	0.354	0.217
fifo	0.348	0.220
fairness_weighted	0.350	0.212
random	0.340	0.203

What the data supports (honestly):

- The *only* paired-significant service gain is **hybrid** > **random** (sev-wt $+0.0125^{\star}$, raw served $+0.0136^{\star}$): you need triage + welfare scoring.
- Against the other *reasonable* policies (greedy, FIFO, fairness) the deltas are **within noise** — under hard saturation, *reachability and capacity bind, not the allocation rule.*
- The HITL gate costs a small, significant $-0.0086^{\star}$dispatch-SLA — the reserve trade.

* = clears the 2-SE screen. Coordination's value here is beating *no* coordination, plus safety and graceful degradation under stress — not a separation between heuristics.

11 Safety & governance

Human-in-the-loop

- gated by explicit, auditable rules: last scarce reserve, large commit (forced-evacuation reason defined & tested, not yet triggered)
- the human gets a **decision package** (context, options, recommendation, risk) — not chatter

Rollback

- reverse a reversible award (lease reclaim); **never yank an on-scene crew** (guard enforced + tested)

Audit & abuse cases

- immutable event log with `trace_id`; read-only auditor
- red-team stress battery: **all 7 scenarios pass** with zero safety-invariant violations
- Sybil, report-storm, comms blackout, 25% asset loss, gauge outage, compound crisis

Stance

- safety constraints are *enforced, not learned*

12 MARL bridge — is multi-agent RL appropriate?

Not for the live system:

- non-stationarity (agents co-learning)
- contested reward (whose objective?)
- governance needs *deterministic, auditable* decisions, not a black box

Yes, offline, for routing: sequential, repeated, simulatable, delayed-reward.

The study (tabular Q-learning, numpy; CTDE via parameter sharing) reproduces **reward hacking** — the Assignment 2 bridge:

policy	true return	flood exposure
risk-aware (RL)	**2.71**	**0.617**
naive (RL)	2.53	0.650
heuristic	2.36	0.674

The naive (travel-time proxy) reward loses on the *true* objective; the learned risk-aware policy even beats the heuristic. **Advisory only** — the heuristic remains the authority.

13 Reflection — honest failures (the rubric rewards them)

- **The fairness lever.** First a silent no-op (deficit scaled every bid equally), then "backfiring," then — after fixing a measurement bug — *within noise*. Kept as an evaluated, documented negative rather than deleted.
- **A served-count bug, caught and corrected.** The read-only auditor double-credited "lives served" when two genuine incidents fused at the COP. Fixing it *lowered* the headline numbers and collapsed most significance — we report the corrected, weaker result rather than the flattering one.
- **The capacity-bound finding.** In a saturated flood the binding constraint is *which barangays you can reach with the assets you have*, not the allocation rule — so the reasonable policies are statistically indistinguishable. That null *is* the finding.

Honest failure analysis, not a victory lap — the full build log is in `docs/REFLECTION.md`.

14 The artifact — reproducible end to end

Engineering

- 39 modules, acyclic DAG, complexity ≤ 10
- 69 behaviour/invariant tests, 97% coverage
- ruff + mypy + markdownlint clean; live loop deterministic
- optional offline LLM perception layer — opt-in, guardrailed, off by default

Run it

- `cli run / eval / stress / rl-train`
- `evals/run_evals.py` — Tier-A gate (non-zero exit)

Where the detail lives

- `README.md` — narrative + rubric map
- `docs/` — 8 design documents
- `paper/report.pdf` — arXiv-style write-up
- `AGENTS.md` — the roster contract
- `evidence/` — every number traces here

One-line defense

A small but credible MAS with a *real* coordination mechanism, a *real* safety story, and *honest* failure analysis — graded on behaviour, reproducible from a single seed.